

Table 4A. Evidence across GREmLN's top-25 candidate regulators

Position	Candidate regulator	CRISPRi evidence	Audited literature evidence	Candidate mRNA context	Independent BTLA molecular evidence
1	KIF22	No detected response (0/10)	None	No qualifying transcriptional response	No qualifying association in measured layers
2	EZH2	No detected response (0/10)	Moderate	No qualifying transcriptional response	No qualifying association in measured layers
3	HMGN3	No detected response (0/9)	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
4	HIRIP3	No detected response (0/9)		No qualifying transcriptional response	Phosphosite association: 2 min, 24 h
5	STAT4	No detected response (0/9)		No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
6	SNAPC4	Not represented in screen	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
7	RBPJ	No detected response (0/9)	Weak	No qualifying transcriptional response	No qualifying association in measured layers
8	CREM ^S	No detected response (0/9)	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
9	STAU2	No detected response (0/9)	None	No qualifying transcriptional response	Phosphosite association: 20 min, 4 h, 24 h
10	MLX	No detected response (0/9)		No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
11	STAT1 ^S	No detected response (0/9)		No qualifying transcriptional response	No qualifying association in measured layers
12	AHR	Detected — BTLA-anti-concordant (7/9)	Moderate	Absent from transcriptomic assay universe	Not measured in protein or phosphoproteomic assays
13	NFIL3	Not represented in screen	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
14	SRP9	Detected — BTLA-concordant (1/8)	None	No qualifying transcriptional response	No qualifying association in measured layers
15	MSC	No detected response (0/8)		Absent from transcriptomic assay universe	Not measured in protein or phosphoproteomic assays
16	GTF3C2	Not represented in screen	None	No qualifying transcriptional response	Phosphosite association: 4 h
17	ZBTB32 ^S	No detected response (0/8)	Weak	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
18	TFDP1	No detected response (0/8)	None	No qualifying transcriptional response	Phosphosite association: 4 h
19	BCL3	No detected response (0/8)	Moderate	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
20	ZNF706	No detected response (0/8)	None	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
21	ZNF121	No detected response (0/8)		No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
22	JUNB ^S	No detected response (0/8)	Moderate	TCR activation 1: 1 h, 24 h	Not measured in protein or phosphoproteomic assays
23	PRDM1 ^S	Not represented in screen	Strong	TCR activation 1: 4 h	Not measured in protein or phosphoproteomic assays
24	IRF8 ^S	Detected — BTLA-concordant (1/8)	Weak	No qualifying transcriptional response	Not measured in protein or phosphoproteomic assays
25	PLAGL2 ^S	Not represented in screen	None	No qualifying transcriptional response	No qualifying association in measured layers

S, shared between the GREmLN and GENIE3 top-25 lists.

Candidate regulators were restricted to the pre-specified pySCENIC human transcription-factor/regulator list.

CRISPRi evidence is model-specific because each model can nominate a different target set for the same candidate. The (confirmed/predicted) denominator uses the native predicted-target budget (not a matched top-N budget). Not represented in screen and failed on-target quality control are shown explicitly and are not converted to zero response.

Audited literature evidence uses the Paperclip v2 TCR-inhibition audit (fixed query "<TF> TCR inhibition"). Strong and moderate literature tiers are audited provisional annotations pending documented human verification.

Candidate mRNA context is contextual evidence only (TCR versus unstimulated control). No candidate showed a qualifying incremental BTLA+TCR versus TCR mRNA response; that contrast is therefore omitted from the column.

Independent BTLA molecular evidence is restricted to directly measured qualifying protein abundance or phosphosite change (BTLA+TCR versus TCR) at the source-native thresholds. Inferred regulon activity, kinase activity, BIONIC, co-IP and synapse annotations are not shown. Molecular direction is not interpreted as supportive or opposing because both model rankings are unsigned. Incomplete assay coverage is reported as not measured and is not negative evidence.

Pale green marks detected CRISPRi response, Strong literature, or qualifying independent molecular association. Pale yellow marks qualifying contextual mRNA. Light grey marks not represented, failed QC, not measured, or None.